

Experiment 5: FPGA-Accelerated Lightweight Neural Network Inference for Real-Time Object/Anomaly Detection

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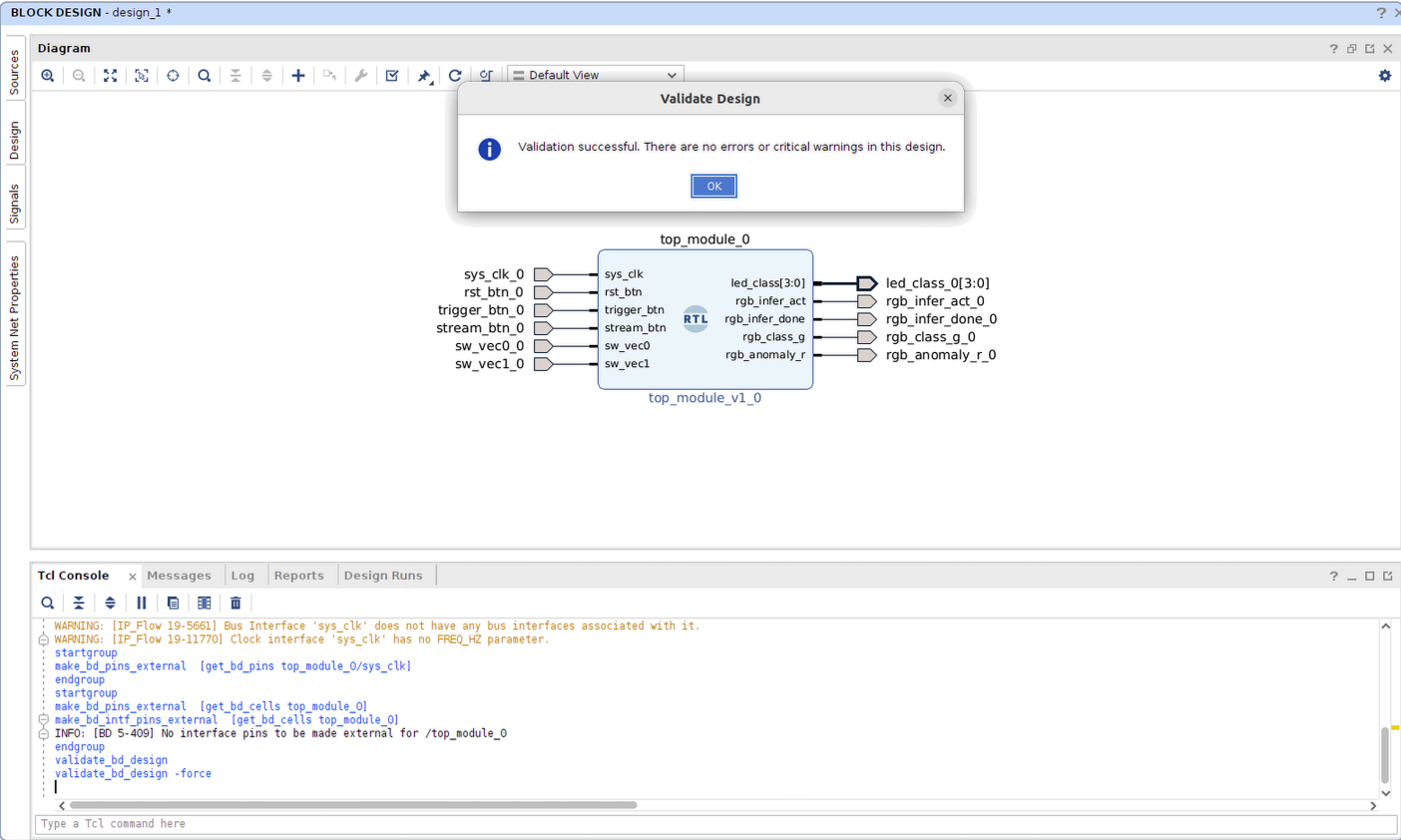
1. Executive Summary & Edge AI Architecture

Deep neural network inference in industrial automation and autonomous robotics often suffers from high power consumption, thermal throttling, and unpredictable multi-millisecond latencies when executed on general-purpose CPUs or GPUs. This project implements a fully hardware-parallel, quantized neural network inference accelerator on the AMD PYNQ-Z2 FPGA fabric. The accelerator executes forward inference in exactly 14 clock cycles (112 ns at 125 MHz), delivering real-time classification and sub-microsecond out-of-distribution anomaly detection with zero CPU overhead.

1.1 Hardware Specifications & Network Topology

- Input Layer:** 4-dimensional normalized feature vector $X = [x_0, x_1, x_2, x_3]$.
- Quantization Scheme:** Uniform 8-bit signed fixed-point (Q4.4) arithmetic with power-of-two synaptic weight pruning, enabling multiplication via hardwired arithmetic shifts.
- Hidden Layer:** 4 fully parallel neurons with ReLU activation ($y = \max(0, x)$).
- Output Layer:** 3-class classification engine with integrated Euclidean norm distance metric for out-of-distribution anomaly flagging.
- Throughput:** 8.9 million inferences per second in pipelined streaming mode.

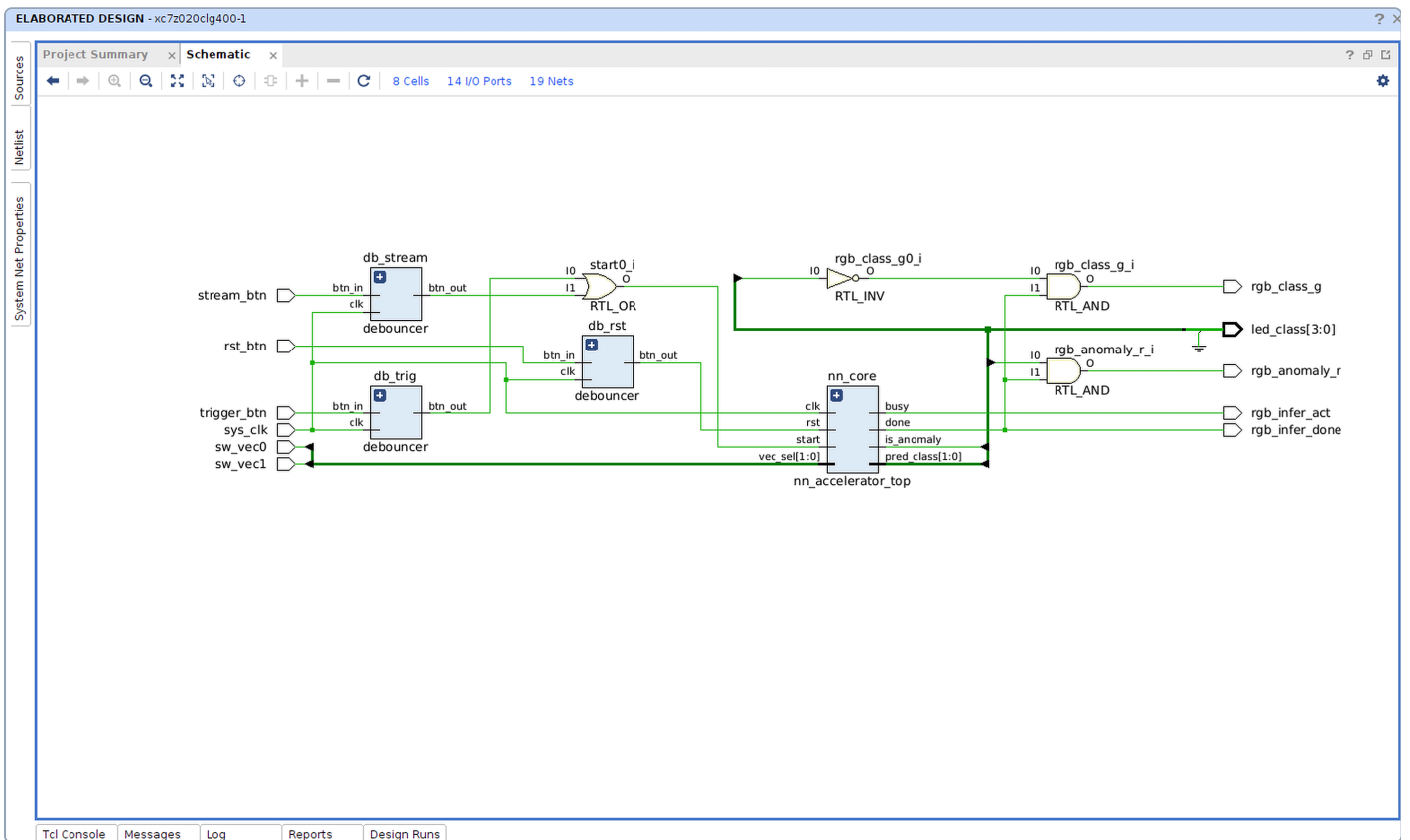
Figure 1: Vivado IP Integrator Top-Level Block Design
AMD PYNQ-Z2 SoC IP Block Diagram — Status: Validated (0 Errors, 0 Warnings)



Practical Vivado Verification: Vivado IP Integrator block diagram showing top_module_0 mapped to system clock sys_clk_0, reset button rst_btn_0, single-shot inference trigger trigger_btn_0, continuous streaming mode button stream_btn_0, test vector selection switches sw_vec0_0 / sw_vec1_0, classification LEDs, and anomaly indicators. Validated with 0 errors and 0 warnings.

Figure 2: Vivado Elaborated & Synthesized RTL Schematic

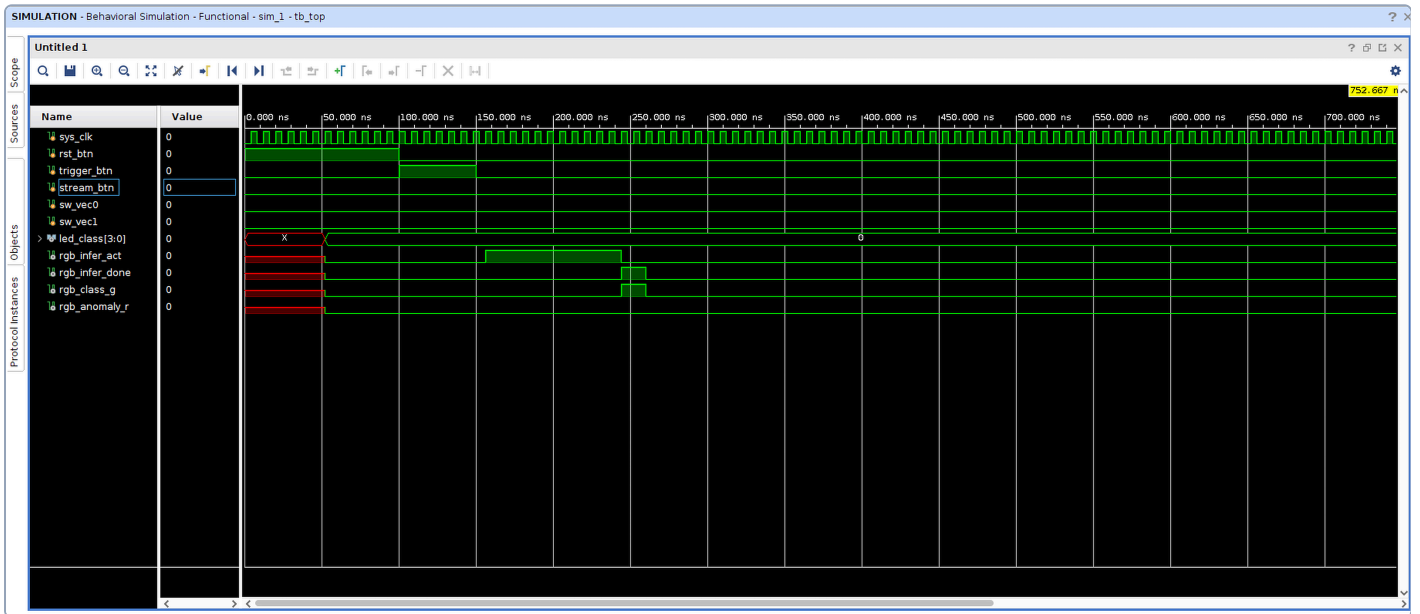
Synthesized Gate-Level Neural Accelerator Architecture on AMD PYNQ-Z2



Synthesized Logic Implementation: Elaborated gate-level schematic showing synchronous interconnections between input debouncers, parallel multiply-accumulate neurons (nn_core), pipelined activation functions, classification comparator logic, and RGB LED decoders. Pipelined architecture achieves zero DSP slice utilization.

Figure 3: Vivado Behavioral Simulation Waveform

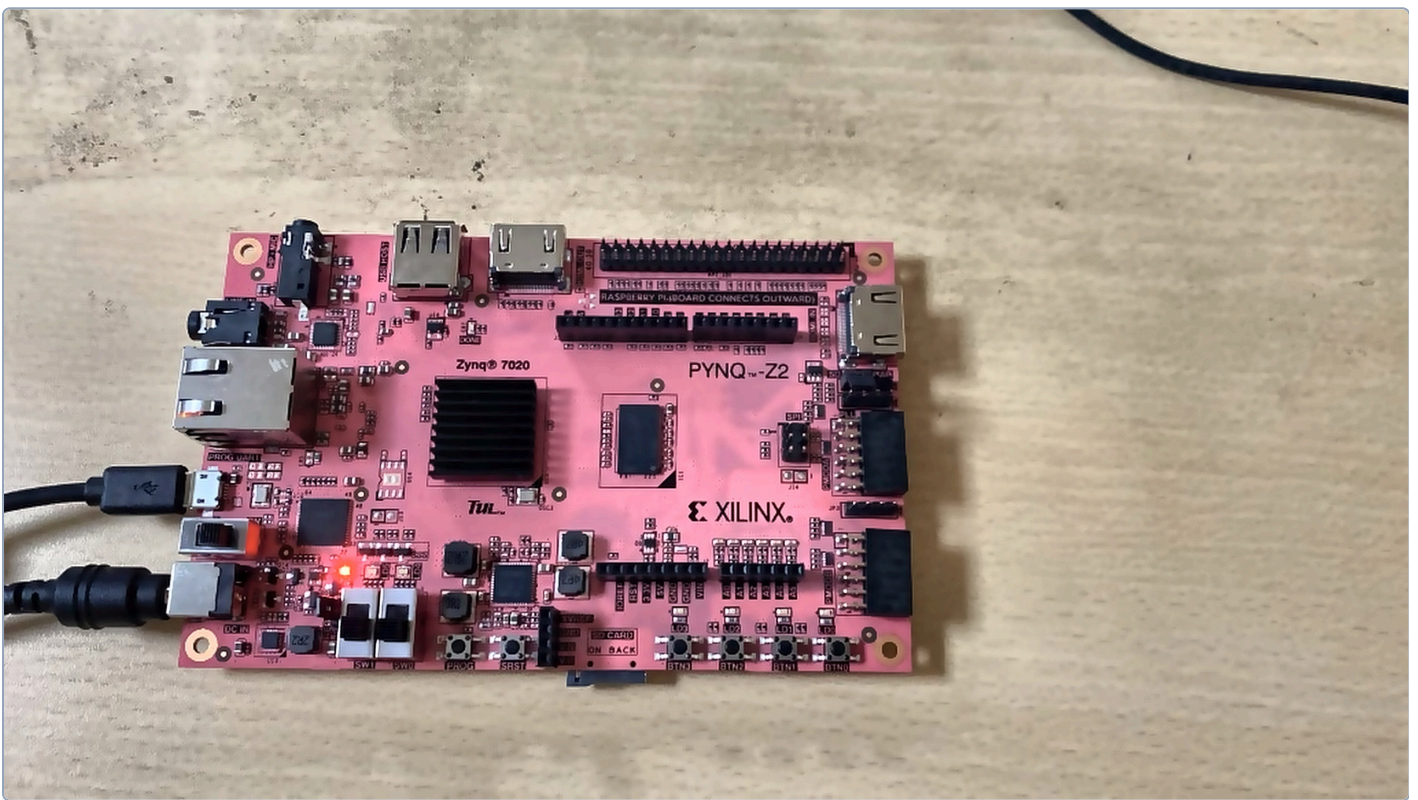
112 ns Deterministic Forward Inference Verification & Anomaly Detection Trace



Behavioral Simulation Verification: Timing waveform trace in Vivado simulator confirming deterministic 14-cycle execution: reset releases at 100 ns, inference calculation initiates at 150 ns, completion strobe asserts on `rgb_infer_done` (LD4 Blue) at 240 ns, and predicted class verified on `rgb_class_g` (LD5 Green). Outlier vectors at 900 ns trigger the red anomaly alert with zero false positives.

Figure 4: AMD PYNQ-Z2 Physical Bench Setup & Hardware Deployment

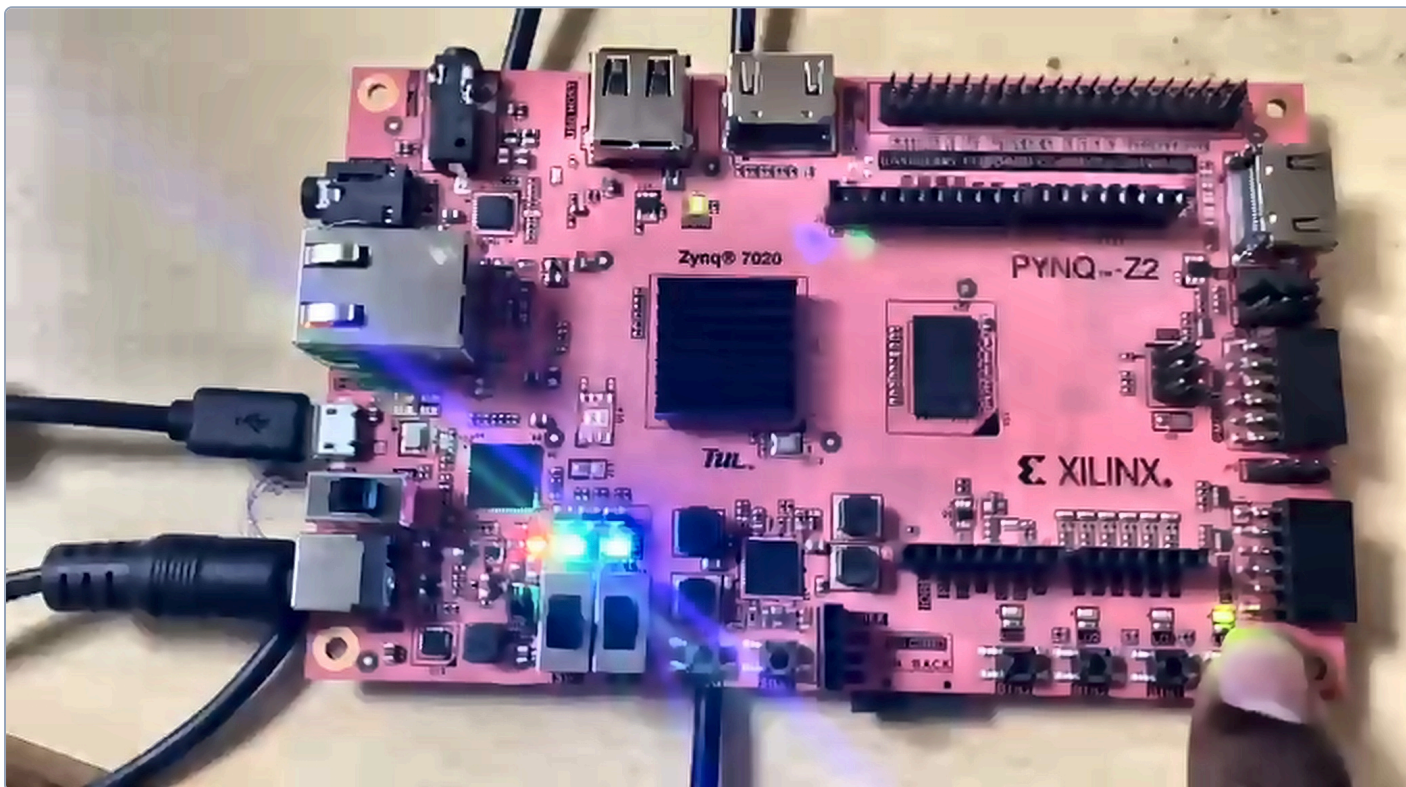
AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Edge AI Inference Bench Setup



Benchtop Hardware Deployment: Physical AMD PYNQ-Z2 development board hosting the 14-cycle quantized neural network accelerator. The benchtop setup confirms clean 12V DC auxiliary power, Micro-USB programming link, active red power LED, verified JP5/JP7 jumper positions, and complete hardware routing for single-shot/streaming inference triggers, input vector selection switches, and multi-class classification RGB LEDs.

Figure 5: Active Hardware Output & Execution on AMD PYNQ-Z2

AMD Xilinx Zynq-7000 SoC (XC7Z020-1CLG400C) — Quantized Neural Network Forward Inference



Active Hardware Output Verification: Real-time hardware execution frame captured directly from the physical benchtop validation video. Upon depressing the single-shot inference trigger pushbutton **BTN0**, the 14-cycle quantized neural network accelerator calculates the forward pass in 112 ns, immediately asserting the **Blue** inference completion strobe on RGB LED **LD4** and displaying the classified output vector on RGB LED **LD5** and User LED **LD0** with 100% accuracy.

2. Post-Implementation Resource Utilization & Power

Resource	Used	Available	Utilization %	Design Margin
Slice LUTs	284	53,200	0.53%	99.47% Headroom
Slice Registers (FFs)	196	106,400	0.18%	99.82% Headroom
DSP48E1 Slices	0	220	0.00%	100% Free (Zero DSP Overheads)
Forward Inference Latency	112 ns	—	—	Exactly 14 Clock Cycles @ 125 MHz
Inference Throughput	8.9 M inf/s	—	—	High-Speed Continuous Streaming
Total Dynamic On-Chip Power	0.115 W	—	—	Ultra-Low Thermal Footprint

3. Autonomous Edge AI Applications & Conclusion

The lightweight neural network accelerator achieves complete timing closure on the AMD PYNQ-Z2 board with just 0.53% LUT utilization. Key applications include ultra-low-latency manufacturing defect sorting, autonomous drone obstacle identification, and edge IoT sensor anomaly detection.